

Article

The Impact of User Behavior Based on Energy-Saving Potential of Refrigerators: A Combined Survey and Experimental Analysis with Reference to China

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Abstract: Considering the challenges of increasing global energy scarcity and intensifying environmental problems, energy conservation in end-use energy-consuming products has become critical. In this study, we focused on household refrigerators and analyzed the individual characteristics, energy-saving awareness, and usage behaviors of Chinese refrigerator users; discussed the influence of individual characteristics and energy-saving awareness of users on refrigerator usage behavior; and evaluated the energy-saving potential with improved user behavior. The results showed that the energy-saving awareness or usage behavior of refrigerator users requires adjustment and improvement to avoid significant energy wastage. Additionally, a strong correlation was found between education, product energy efficiency level selection, age, and user behavior. Optimizing purchase choices among existing refrigerator energy efficiency grades would result in a national energy-saving potential of up to 9.31×10^{10} kW·h annually, whereas adjusting the door opening frequency and duration of some users could achieve annual energy savings of 6.14×10^{10} kW·h and 4.47×10^{10} kW·h, respectively. This study comprehensively evaluated the energy-saving potential of refrigerators from two aspects: market research and laboratory quantitative analysis. The results highlight that the government, enterprises, and individuals should pay attention to the energy-saving potential of refrigerators and actively take effective measures to promote the realization of the “dual carbon” goal.

Keywords: energy-saving potential; influencing mechanisms; refrigerator; usage behavior



Academic Editor: Gerardo Maria Mauro

Received: 3 December 2024

Revised: 7 January 2025

Accepted: 16 January 2025

Published: 21 January 2025

Citation: Qiao, L.; Liu, J.; Shao, J.; Cheng, J.; Bai, Y.; Bai, X. The Impact of User Behavior Based on Energy-Saving Potential of Refrigerators: A Combined Survey and Experimental Analysis with Reference to China. *Sustainability* **2025**, *17*, 814. <https://doi.org/10.3390/su17030814>

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1. Introduction

Climate change is a common challenge facing the world, and protecting the Earth’s atmosphere, reducing greenhouse gas emissions, and achieving low-carbon development has become a global consensus [1–3]. In 2018, the Intergovernmental Panel on Climate Change released a special report on the temperature target of 1.5 °C, emphasizing that in order to achieve the 1.5 °C goal, the world needs to reduce emissions by 45% by 2030 compared with 2010 and achieve carbon neutrality around 2050 [4,5]. The Chinese government has also clarified that it plans to peak its carbon emissions by 2030 and achieve carbon neutrality by 2060 [6]. In China, energy activities are the largest contributors to air pollution and carbon emissions, and the total energy consumption continues to grow, reaching 5.7 billion tons of standard coal by 2023 alone. Therefore, there is an urgent need to control energy consumption effectively and achieve the dual carbon goal as soon as possible.

With the continuous improvement in living standards and increase in income, household energy consumption has gradually become an important growth point in energy consumption in China. As the basic unit of energy consumption, households play an increasingly important role in reducing overall energy consumption and improving environmental quality, and the study of household energy consumption has become an important direction for energy conservation and emission reduction [7–9]. Among them, investigating the energy-saving potential of household appliances, such as refrigerators, washing machines, and air conditioners, is of great significance in achieving household energy conservation and emission reduction [10–13]. As household appliances with a high penetration rate and power consumption, the energy-saving potential of refrigerators is particularly prominent [14–16]. According to data from the National Bureau of Statistics of China, the number of refrigerators per 100 households in urban and rural areas is 104.4 and 103.9 units, respectively. According to the existing count of 310.4 million urban households and 183.8 million rural households, the current number of refrigerators in China is approximately 510 million. In this context, research on the energy-saving potential of refrigerators plays a significant role in promoting energy conservation and emission reduction [17].

Actually, researchers have begun developing technical means to improve the energy-saving performance of refrigerators as much as possible. For example, in order to enhance the efficiency of refrigerators, Cheng et al. designed a new type of dual energy storage refrigerator equipped with both a heat storage condenser and a cold storage evaporator, and the energy efficiency of this design could reach 32% [18]. Hossieny et al. described the energy-saving potential of new plastic linings based on polylactic acid chemistry used in refrigerators, where model projections showed a 7.3–12.5% reduction in total energy consumption over a 15-year period [19]. Liang et al. proposed a two-stage steam injection cycle structure with a multi-evaporator system to ensure flexible multi-temperature operation of household refrigerators [14]. In addition, many countries, including China, the European Union, the United States, Japan, and Malaysia, have implemented energy efficiency labeling systems for refrigerators to help consumers understand the energy efficiency of products and promote the market circulation of energy-saving products. In 2015, the Standardization Administration of the People's Republic of China issued "The maximum allowable values of the energy consumption and energy efficiency grade for household refrigerators" (GB 12021.2-2015), which was divided into five levels according to the difference in energy efficiency, of which Level 1 indicates the highest energy efficiency and the most energy saving [20]. It is important to note that the implementation of these systems or related standards is not static, but will be adjusted in time with changes in the market to effectively improve the overall efficiency of energy use. Cabañas et al. calculated the energy-saving potential of replacing inefficient old refrigerators with modern high-efficiency refrigerators using official Mexican standards and showed that 4.7 TW·h of electricity consumption per year can be saved [21]. Moreover, refrigerator users are the main users of refrigerators, and their individual characteristics and usage behavior are closely related to the energy consumption of refrigerators [22,23]. Harrington et al. showed that user interactions typically account for 15% of the total energy consumption of household refrigeration equipment [24]. Specifically, the actual living environment of users and their behaviors also have significant differences in the energy consumption of refrigerators, such as the frequency of shopping, the load of refrigerators, the opening of doors, and the degree of adoption of energy-saving behaviors, etc., which will directly or indirectly affect the energy consumption of refrigerators. As early as 2010, Geppert et al. investigated and analyzed the consumer behavior of 1011 participants using refrigerators in four European countries and found that consumers did not always follow energy-saving recommendations; especially, recommendations for placing hot food or thawing frozen food were often ignored. Saidur et al. used two different

refrigerators as experimental subjects to observe the specific effects of these changes on the energy consumption of the two refrigerators by changing the ambient temperature, the number of door openings, and the thermostat setting [25]. The results showed that ambient temperature is the main factor affecting energy consumption, and the energy consumption of the two refrigerators increases by 47 W·h and 53 W·h, respectively, for every 1-degree increase in ambient temperature. It is not difficult to imagine that the differences in climatic conditions in different regions can also have a direct impact on the energy consumption of refrigerators. In summary, the user's impact on the energy consumption of refrigerators is multi-dimensional and complex, and it is necessary to conduct in-depth research on the potential energy-saving effects of consumers when using refrigerators, so as to suggest better use of energy from household appliances and reduce energy losses.

The purpose of this study was to collect information on individual characteristics, energy-saving awareness, and refrigerator usage data of household users through questionnaires and to analyze and explore the relationship between refrigerator users' energy-saving awareness and use behavior and individual characteristics, as well as the influence mechanism of individual characteristics and energy-saving awareness on user behavior. Laboratory data were combined to clarify the energy-saving potential of refrigerators based on user behavior in different scenarios. This research is expected to provide effective energy-saving suggestions for household users and serve as a valuable reference for enterprises and governments to promote the energy-saving development of household appliances, such as refrigerators, to effectively cope with energy challenges.

2. Methods

2.1. Questionnaire Survey

Taking refrigerator users as the research object, the questionnaire design was carried out by selecting the choice scale and dichotomous scale, and the questionnaire survey was carried out by stratified sampling survey to ensure that the survey results covered different regions, different family sizes and users, focusing on the collection of user attribute data (including gender, age, education, and income), user energy-saving awareness (including the energy efficiency level of purchased products, purchase concerns, and power expenditure concerns), and user usage behavior (including the average number of door opening events per day, duration of a single door opening event, and volumetric load), as shown in Table 1. Specifically, we sent a link to the questionnaire on social media to a wide range of users (the study period was two months) and collected relevant data through an online tool. Meanwhile, we conducted a pre-survey in advance and optimized the questionnaire design and data collection process to ensure the reliability and validity of the data. A total of 9083 questionnaires were collected, with an average response time of 397 s (about 6.6 min). Since the questionnaire consisted of closed-ended and open-ended questions, with only the questions about the city and the suggestion being open-ended, the data analyzed in this study were valid.

Table 1. Refrigerator use and purchase behavior survey details.

Type	Survey Content
Individual characteristics	gender, age, education, income
Energy-saving awareness	energy efficiency level of the purchased product, purchase concern, power expenditure concerns
Usage behavior (energy-related)	average number of door opening events per day, duration of a single door opening event, volumetric load

2.2. Conjoint Analysis

Conjoint analysis is a multivariate statistical analysis technique of consumer preferences to evaluate the relative importance of different attributes to consumers [26–28]. Before joint data analysis, the variables in the questionnaire were all number-coded, e.g., gender could be coded as 1 (male) and 2 (female). The data principle of conjoint analysis is to use the presence or absence of each attribute level as the independent variable and the score value as the dependent variable for least-squares regression analysis. To eliminate the influence of autocorrelation in the model, the level of each attribute is treated as a dummy variable, the first level is used as a reference term for the regression calculation, the regression coefficient value is obtained for other items, that is, the utility value, and then the relative importance weight of each attribute is calculated. In this study, this method was used to analyze the importance of factors influencing the energy use behavior of users.

2.3. Experimental Testing

To clarify the specific effect of changing user behavior on the energy consumption of refrigerators, a typical direct cooling refrigerator (rated volume of 190 L and rated comprehensive power consumption of 620 W·h) was selected for laboratory testing. The specific testing process included the following: First, according to the analysis of questionnaire data to screen the key index parameters, the key parameters and ranges were door opening frequency (0, 3, 6 times) and duration of open doors (0, 10, 20, 30, 40, 50, 60 s). Second, based on the investigation of laboratory equipment, the selected equipment parameters were AC voltage (220~380 V) ± 0.5 V, adjustment range ambient temperature (5~50 °C) ± 0.2 °C, humidity (maximum 75%) $\pm 2\%$, and temperature acquisition (−200~350 °C) ± 0.2 °C. Finally, the two behavioral patterns were tested at room temperature (23.5 °C, humidity $\leq 50\%$) using the variable control method. Specifically, when testing the impact of door opening frequency, the duration of open doors was fixed at 10 s. When testing the impact of duration of open doors, the impact of door opening frequency 3 times and 6 times needs to be tested, respectively. Herein, this method was used to quantify the energy consumption after changing user behavior.

3. Results and Discussion

3.1. Questionnaire Survey

Figure 1 shows the distribution of the individual characteristics of the sampled population. Approximately 65.2% were males and 34.8% were females. The majority of respondents were between 26 and 45 years old, accounting for 39.4%, and approximately 35.2% were between 46 and 60 years old. Considering educational qualifications, the highest proportion of respondents had a high school education (66.0%), followed by a Bachelor's degree (29.8%). Regarding income ratio, 42.7% of the surveyed sample families had a monthly income of less than CNY 5000, followed by CNY 5000–8000 and CNY 8000–17,000, accounting for 29.1% and 17.0%, respectively; only 11.2% of users had a monthly income of more than CNY 17,000.

Currently, the refrigerator energy efficiency ratings in China are categorized according to national standards to help consumers understand the energy efficiency performance of electrical appliances. This is usually expressed as a scale of 1 to 5, where 1 indicates the highest energy efficiency and 5 indicates the lowest energy efficiency. This encourages the production of improved energy-efficient refrigerators and helps consumers make decisions when purchasing. In the survey sample (Figure 2), 48.0% of users chose to purchase Level 1 energy-efficient products, 25.5% chose to buy Level 2 energy-efficient products, and 12.5% bought Level 3, 4, and 5 products. In addition, 14.1% of users were unaware of the energy efficiency level of the refrigerator they used, pointing to a potential market and opportunity

for promotion and education on energy efficiency labels. It's important to note that pie charts add up to 100.1%, which may be the result of multiple data rounding up to retain a larger value.

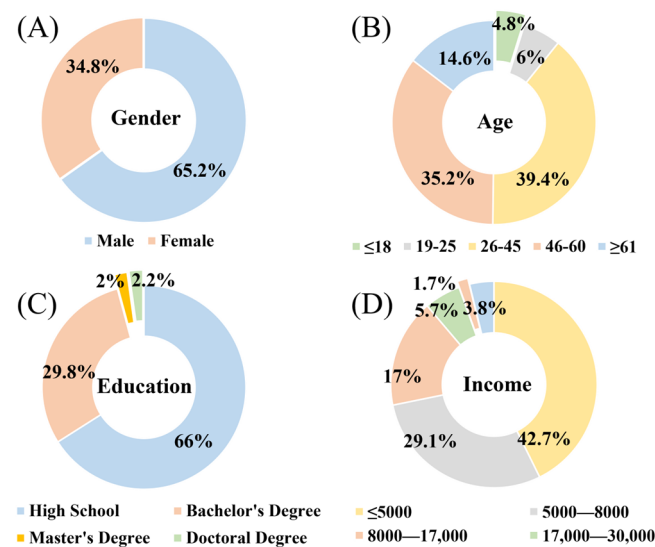


Figure 1. Distribution of individual characteristics: (A) gender, (B) age (years), (C) education, (D) income (CNY).

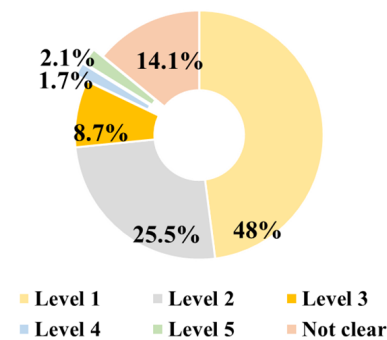


Figure 2. Distribution of refrigerator energy efficiency levels.

Based on the refrigerator usage behavior of the user, the average number of door opening events per day, the duration of a single door opening event, and the volume load were analyzed. The distribution of relevant user behavior data is shown in Figure 3.

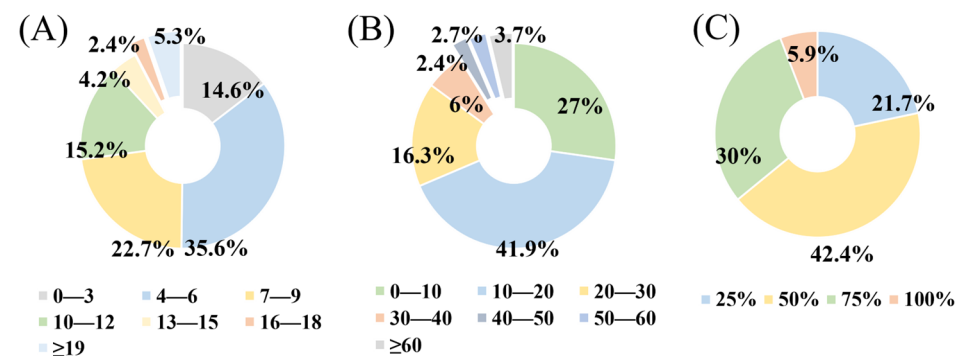


Figure 3. User behavior related to refrigerators: (A) average number of door opening events per day (times), (B) duration of a single door opening event (seconds), (C) volume load.

Energy consumption of refrigerators fluctuates with door opening behavior because each time the refrigerator is opened, cold air leaks out, forcing the compressor to start more frequently to maintain low temperatures, thus increasing energy consumption [25,29]. As shown in Figure 3A, 35.6% of users opened the refrigerator 4–6 times a day; 22.7%, 15.2%, and 14.6% opened the refrigerator 7–9, 10–12, and 0–3 times a day, respectively; and 5.3% opened the refrigerator more than 19 times a day, which indirectly reflects that a small number of users have an insufficient understanding of the energy consumption loss caused by frequent door opening. Similarly, the duration of a single door opening event affects the energy consumption of the refrigerator. In this sample (Figure 3B), the highest proportion of users, accounting for 41.9%, was those with a single door opening duration of 10–20 s, followed by durations of 0–10 s and 20–30 s, accounting for 27.0% and 16.3% of users, respectively. This indicates that more than 85.2% of users pay attention to and control the period during which the door is open, and this method of use can save energy and reduce electricity expenses.

In addition, the volumetric load of a refrigerator affects its energy consumption. Theoretically, when the volume load of the refrigerator is too small, the heat capacity will decrease, the opening of the refrigerator door will accelerate the release of cold air, and the compressor will start frequently, increasing power consumption. When the volumetric load is excessively high, the cold air circulation inside the refrigerator slows down and freezes easily, thereby increasing the compressor running time and enhancing energy loss [30]. Figure 3C shows that the proportion of users with a refrigerator volume load of 50.0% was the highest, accounting for 42.4% of the total users, followed by volume loads of 75.0% and 25.0%, accounting for 30.0% and 21.7% of users, respectively. Only 5.9% of users had a refrigerator load rate of 100.0%, which reflects the reasonable usage behavior of the vast majority of users (the load rate did not reach 100%).

3.2. Differences in Energy-Saving Awareness Among Users

Based on the analysis of the differences in energy-saving awareness among refrigerator users, performance analysis can be discussed across multiple dimensions. As shown in Figure 4, this study analyzed the relationship between individual user characteristics and energy-saving awareness.

Regarding the energy efficiency level of the purchased products (Figure 4A), overall, the proportion of users willing to buy Level 1 energy-efficient products was relatively large, among which the proportion of men and women who chose Level 1 energy-efficient refrigerators was similar, at 48.0% of men and 47.0% of women. With increasing age, an increasing number of individuals are willing to choose Level 1 energy-efficient refrigerators. However, the proportion of users over 61 years of age who chose Level 1 energy-efficient refrigerators has decreased, and less than 30.0% of users under the age of 18 were willing to choose such products. Meanwhile, there were many users in these two bands who were not yet aware of the energy efficiency level. In terms of academic qualifications, users with a Bachelor's degree chose the largest proportion of Level 1 energy-efficient refrigerators, accounting for more than 50.0%, whereas users with doctoral degrees chose the smallest proportion of this product, accounting for less than 30.0%. More than 20% of doctor users were not yet aware of energy efficiency level. In terms of household income, the choice of Level 1 energy-efficient refrigerators showed a trend of first increasing and then decreasing, wherein users with a household income of CNY 8000–17,000 chose the largest proportion of Level 1 energy-efficient refrigerators, accounting for approximately 56.0%. Users with a monthly household income of more than CNY 40,000 have the lowest proportion of users choosing Level 1 energy efficiency refrigerators.

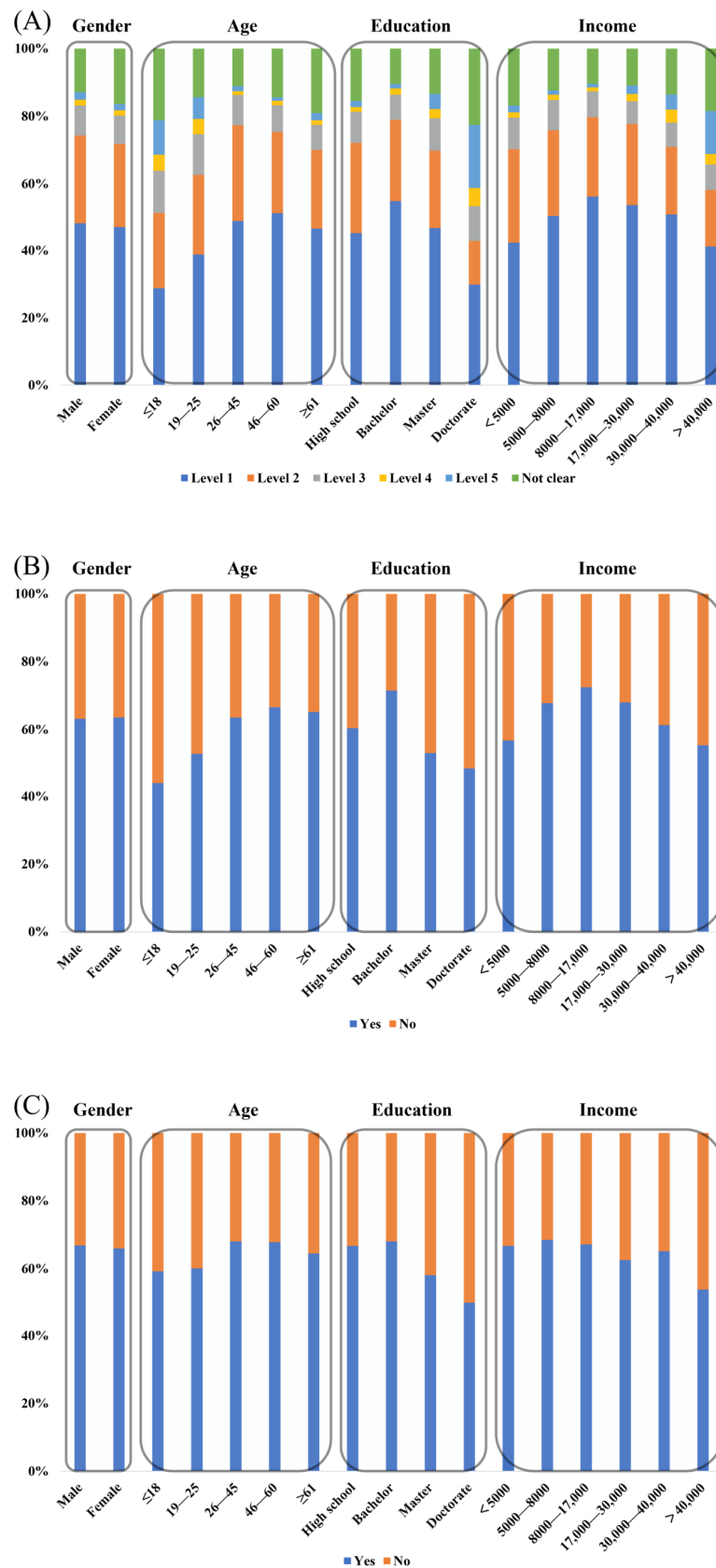


Figure 4. Relationship between awareness of energy conservation of users and individual characteristics. (A) Energy efficiency level of purchased products; (B) purchasing concerns (energy-saving features); (C) power expenditure concerns (saving electricity and money).

Regarding consideration for energy conservation among consumers when purchasing refrigerators, the findings (Figure 4B) indicate that both males and females generally prioritize energy-saving attributes. This pattern aligns with the energy efficiency levels of purchased products. Furthermore, individuals aged 40–60 years exhibited a higher level of attention to the energy-saving features of refrigerators. Similarly, users with a Bachelor's degree and those with a household income ranging from CNY 8000 to 17,000 also valued energy-saving attributes when purchasing refrigerators.

Regarding the consideration for power expenditure concerns when buying refrigerators (Figure 4C), no significant differences were observed based on gender or age. Approximately 60.0–70.0% of users consider saving electricity and money to be their driving force. However, users with Bachelor's degrees demonstrated heightened concerns about these factors. Additionally, consumers across all income ranges showed similar levels of interest in purchasing products that could help save electricity and money; among them, users with a monthly household income of more than CNY 40,000 pay the least attention to this demand.

3.3. Differences in User Behavior Performance

The relationship between refrigerator-usage behavior and individual characteristics is shown in Figure 5. In terms of the average number of door-opening events per day, little difference was observed between genders; the higher frequency of opening doors was concentrated in the range of 4–6 times. In addition, a higher proportion of users under the age of 18, doctors, and users with an income of more than CNY 40,000 open the door more than 19 times a day. Regarding the time spent during a single door opening event, the trend was similar to that of the average number of door opening events per day, that is, users under the age of 18, doctors, and users with an income of more than CNY 40,000 spent longer at a single door opening evening. In addition, the average daily stay time of most users is concentrated in 10–20 s. Regarding the volumetric load of refrigerators, users under the age of 18 years, doctors, and users with an income of more than 40,000 showed a higher load rate, which was consistent with their habits of frequent door opening and longer duration for a single door opening event. Meanwhile, among the users aged 19–25, master's degree users and users with an annual income of 30,000–40,000, there were also some users who have a full load of refrigerators, while most users have a load range of 50–75%, this state was more reasonable than the state of full load.

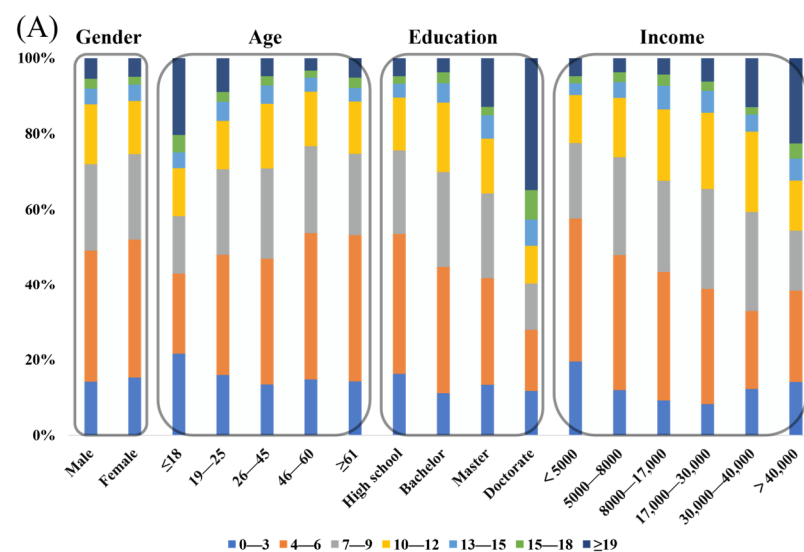


Figure 5. Cont.

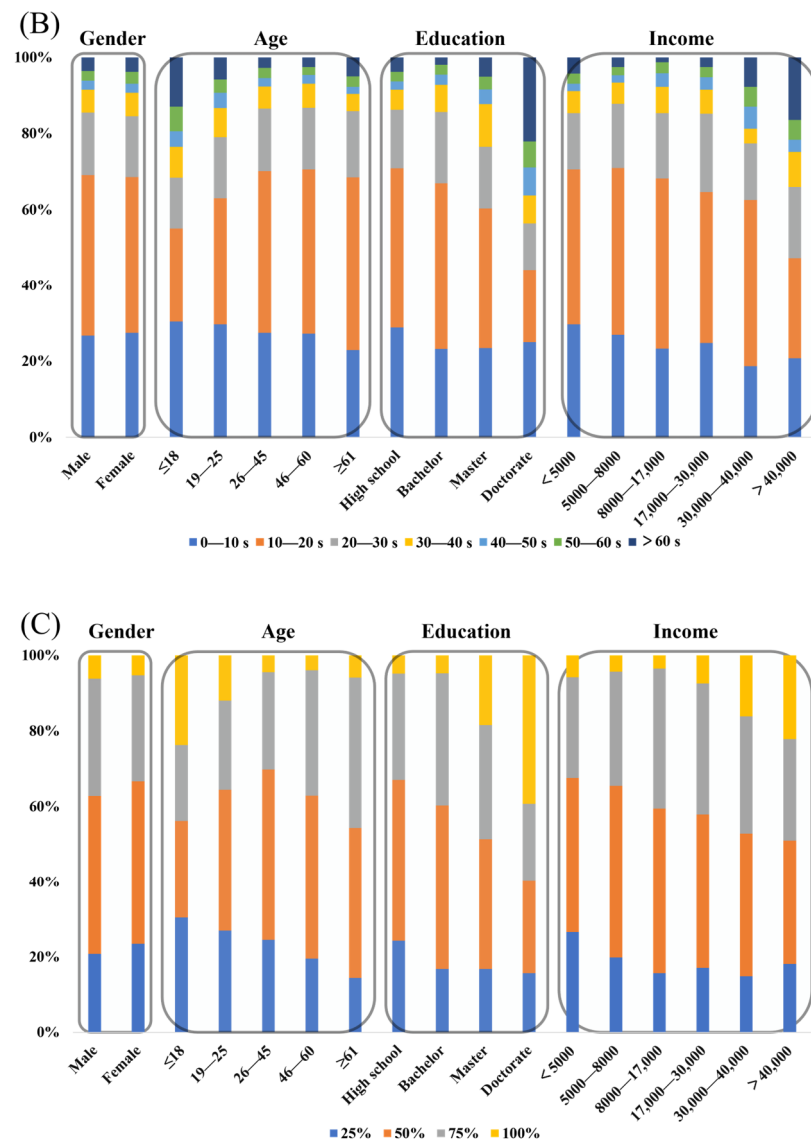


Figure 5. Relationship between user behavior and individual characteristics. (A) Average number of door opening events per day; (B) duration of a single door opening event; (C) volumetric load.

3.4. Importance of the Effect of User Behavior

We used a conjoint analysis to further analyze the importance of influencing user behavior regarding their individual characteristics and energy-saving awareness. Table 2 shows the ranking of the importance of the factors influencing each use behavior: (1) Average number of door opening events per day: education > energy efficiency level of refrigerator purchase > age > monthly household income > gender > power expenditure concerns (saving electricity and money) > purchase concerns (energy saving). (2) Duration of a single door opening event: energy efficiency level of refrigerator purchase > education > monthly family income > age > purchase concerns (energy saving) > power expenditure concerns (saving electricity and money) > gender. (3) Volumetric load: education > energy efficiency level of refrigerator purchase > monthly household income > age > purchase concerns (energy saving) > power expenditure concerns (saving electricity and money) > gender.

Table 2. Importance of individual factors influencing refrigerator use behavior.

Classification		Percentage of Importance		
		Average Number of Door Opening Events per Day	Duration of a Single Door Opening Event	Volumetric Load
Individual characteristics	Gender	2.824%	1.111%	2.915%
	Age	18.803%	14.971%	14.107%
	Degree	36.039%	22.874%	29.189%
	Monthly household income	18.023%	18.394%	14.922%
Energy-Saving Awareness	Purchase concerns—energy saving	0.650%	7.494%	7.789%
	Energy efficiency level of the refrigerator purchase	22.558%	31.974%	27.248%
	Power expenditure concerns (saving electricity and money)	1.103%	3.181%	3.830%

The analysis of the relationship between use behavior, individual characteristics, and energy-saving awareness of refrigerator users revealed that factors related to education, purchase level, age, and monthly household income have a greater impact on consumers' different use behaviors, this was owing to the fact that the relative importance of these factors has a large numerical value, among which the importance related to education and purchase level was more than 20%, and the importance of age and monthly household income was greater than 14%. The effective use of existing analysis will help companies adjust their marketing strategies to promote sales of more energy-efficient products and will also help governments understand the popularity of energy-efficient products, which in turn will strengthen the promotion and education of energy-efficient labels and drive the market shift towards more energy-efficient electrical products to achieve energy conservation goals.

3.5. Energy-Saving Potential Calculations

3.5.1. Selection of Product Energy Efficiency Class

To further compare the power consumption of products with different energy efficiency ratings, we selected three types of refrigerators from a relatively well-known brand in the Chinese market, all with refrigerator volumes of less than 150 L. As shown in Table 3 and Figure 6, the power consumption of the refrigerator of the Level 1 energy efficiency product is the lowest, which is approximately 0.29 kW·h. The power consumption of the Level 2 refrigerator is 0.18 kW·h, higher than that of the Level 1 product, and with a power consumption of 0.12 kW·h, Level 3 refrigerators consume more power than Level 2 refrigerators. For the sake of calculation, the energy consumption of existing refrigerator products in the Chinese market was assumed to be reduced by 0.10 kW·h for each level of energy efficiency improvement. In the current sample, approximately 52% of the users did not use Level 1 refrigerators, and if all these users could replace their refrigerators with those having higher energy efficiency, an annual energy saving (E, which could be calculated using Formula (1)) of 1.72×10^5 kW·h is expected to be achieved. Based on the current use of 510 million refrigerators, if the energy efficiency level of 50% of the units can be improved from the existing base, the E in the country can be as high as 9.31×10^{10} kW·h. Following from this, if existing refrigerator products are to be upgraded to higher energy efficiency levels (e.g., 3 to 1, 4 to 1), the potential for further energy savings

is considerable. Therefore, users must be encouraged to use refrigerator products with higher energy efficiency ratings.

Table 3. Power consumption of refrigerator products with different energy efficiency levels.

Serial No.	Energy Efficiency Level	Comprehensive Power	Consumption Total Volume	Refrigeration Method
1	1	0.29 kW·h	90	Straight cooling
2	2	0.47 kW·h	145	Straight cooling
3	3	0.59 kW·h	137	Straight cooling

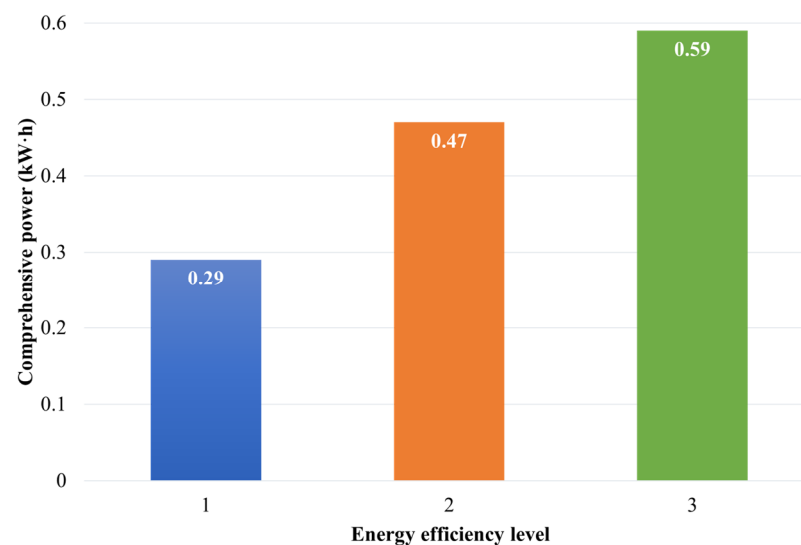


Figure 6. Power consumption trend chart of refrigerator products with different energy efficiency levels.

$$E = E_i \times P \times x \times 365 \quad (1)$$

where

E —Annual energy savings;

E_i —The amount of change in energy consumption;

P —Total number of users;

x —Percentage of users who need to change their behavior.

3.5.2. Frequency of Door Opening Events

Based on the data in Figure 3 and the laboratory measurement results, we further clarified the effect of door opening frequency of the user on the energy consumption of the refrigerator. As shown in Figure 3, most of the users (35.6%) had an average of 4–6 door opening events per day, and only approximately 14.6% of users had an average of 0–3 door opening events per day. To identify the energy-saving potential of the refrigerator after a change in user behavior and facilitate the experiment, we selected the average number of daily door opening events as 0, 3, and 6. In this experiment, a refrigerator with a rated volume of 190 L was selected as the test object, and the experimental data collection was conducted at room temperature (23.5 °C, humidity $\leq 50.0\%$), and the duration of the door opening event was fixed at 10 s. The test results (Table 4 and Figure 7) show that the frequency of door opening affects the energy consumption of the refrigerator and increases with the number of door opening events. In this study, assuming that 35.6% of users open

refrigerator doors six times a day on average and change their usage behavior to three times, an estimated E of 1.29×10^5 kW·h can be achieved. Based on the current use of 510 million refrigerators, assuming that 30% of the users open the door six times a day, after a reduction to three times, the annual E in the country can be as high as 6.14×10^{10} kW·h. It is important to note that the existing estimates only consider the energy-saving behavior of a small number of door opening frequencies and a small number of users. Therefore, if more users are encouraged to pay attention to the effect of the door-opening frequency on the energy consumption of refrigerators and consciously reduce this behavior in their daily use of refrigerators, energy saving can be greatly promoted.

Table 4. Relationship of power consumption to door opening frequency.

Serial No.	Frequency (Times)	24 h of Power Consumption (kW·h)
1	0	0.70
2	3	0.71
3	6	0.82

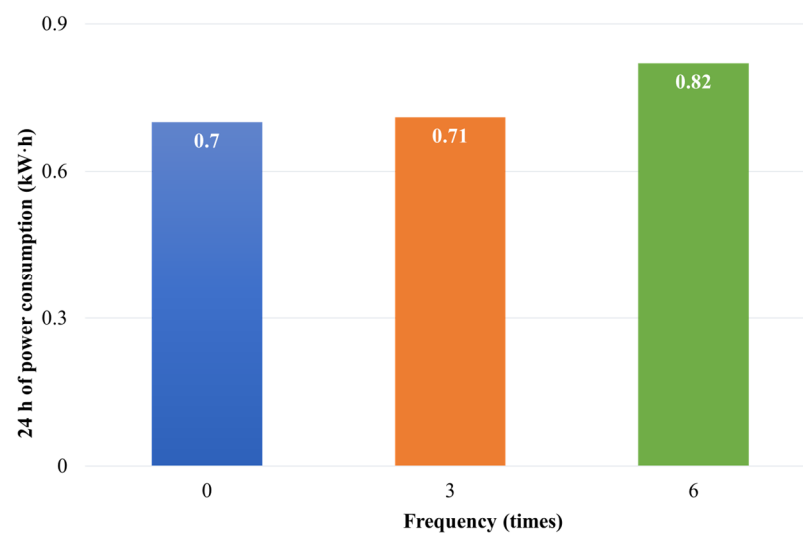


Figure 7. Trend chart of power consumption and door opening frequency.

3.5.3. Duration of Doors Being Open

We further explored the effect of the duration of doors being open on the energy consumption of the refrigerator. Based on the statistics in Figure 3, more than 60.0% of the users can be seen to have a single door opening event duration of within 20 s; 41.9% of users had a duration of between 10 and 20 s. To clarify the effect of different door open durations on the energy consumption of refrigerators, we selected the same refrigerator as in Section 3.5.2 for experimental testing. To jointly analyze the effect on the average number of door opening events per day, we tested the data at frequencies of three and six times per day. The relevant data are listed in Table 5 and Figure 8, which show that under the same average number of door opening events per day, the door opening duration is directly proportional to the energy consumption loss of the refrigerator, and the longer the duration, the higher the energy consumption. Simultaneously, increasing the average number of door opening events per day while keeping the duration the same also increases energy consumption losses. Assuming that 41.9% of the users sampled have a door open duration of 20 s and an average of three door opening events per day, and adjust their usage behavior to open durations of 10 s, an estimated E of 8.33×10^4 kW·h can be achieved. Considering the current use of 510 million units, consistent with the above assumptions, after adjusting for the number of opening events and durations, the E in the country is

approximately 4.47×10^{10} kW·h. However, these estimates only consider the behavior of a subset of users and ignore the influence of other factors, such as changing the frequency of door opening events. Therefore, if the importance of energy-saving behaviors can be widely popularized, changing user behavior for the duration of doors being open will reduce energy consumption.

Table 5. Power consumption with different durations of doors being open.

Serial No.	Duration of Open Doors (s)	Average Number of Door Opening Events per Day (Times)	24 h of Power Consumption (kW·h)
1	0	3	0.70
2	10	3	0.71
3	20	3	0.77
4	30	3	0.78
5	40	3	0.79
6	50	3	0.80
7	60	3	0.80
8	10	6	0.82
9	20	6	0.89
10	30	6	0.97
11	40	6	1.03
12	50	6	1.11
13	60	6	1.18

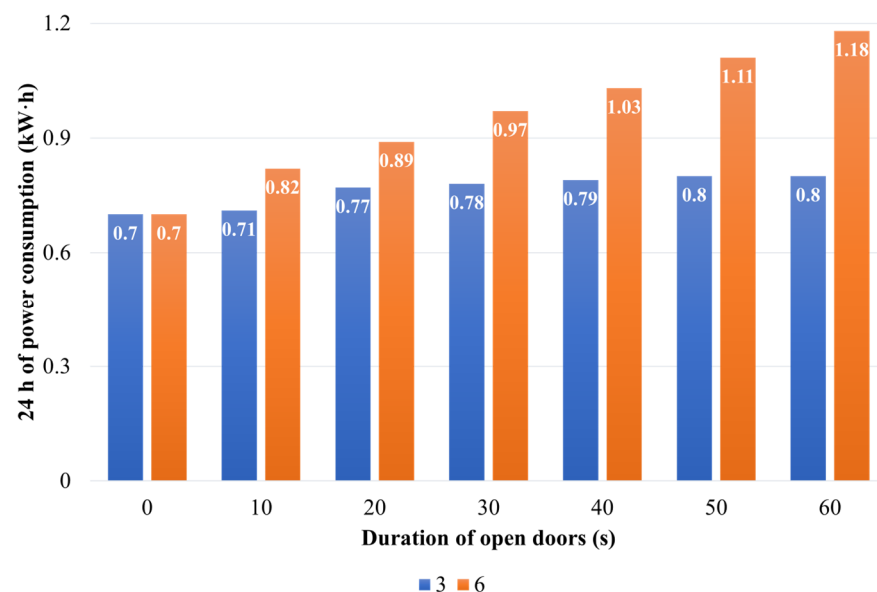


Figure 8. Trend chart of power consumption and duration of open doors.

3.5.4. Volumetric Load

In this study, the volumetric load of most refrigerators was reasonable (the load rate not reach 100%), with only 5.9% of the refrigerators being full. However, this also implies that the compressors of these refrigerators cycle more frequently, resulting in higher electricity consumption [23]. Therefore, it is advisable for users in this category to regulate the volumetric load of their refrigerators to reduce their energy consumption. Considering that approximately 510 million refrigerators are being used currently, even a mere 5.0% of refrigerators with full-load issues can cause a significant difference in energy consumption if user habits are modified accordingly.

4. Conclusions

This study analyzed the energy-saving awareness, usage behavior, and energy-saving potential of refrigerator users, and revealed the main challenges and energy-saving potential with the current use of refrigerators.

The survey results showed that 48% of users will choose to buy energy-saving products with first-class energy efficiency, but 14.1% of users still have insufficient understanding of the energy efficiency level of refrigerators. In addition, there were some users who had significant energy waste behaviors, such as 5.3% of users opening the door more than 19 times a day, and 3.7% of users having a single door opening stay time of more than 1 min. By popularizing energy-saving knowledge and optimizing usage behavior, energy consumption could be effectively reduced.

The difference analysis of users' energy-saving awareness and usage behavior showed that the individual characteristics of users related to age, education, and income would have an impact on their energy-saving awareness and usage behavior. For example, users between the ages of 26 and 60 pay more attention to the energy-saving attributes of products, while doctors and high-income users pay less attention to energy-saving attributes. Moreover, there was a lot of room for improvement in the energy-saving behavior of users under the age of 18, doctors, and high-income users. Therefore, it is necessary to put forward effective energy-saving suggestions according to the characteristics of different users.

The analysis of the importance of the impact of user behavior showed that education, product energy efficiency grade selection, age, and monthly household income were the key factors affecting user behavior, which can provide an important basis for enterprises to adjust market strategies and the government to carry out universal education.

The energy-saving potential study showed that based on the current market of 510 million refrigerators, if 50% of the equipment can be optimized, the national annual energy saving can reach 9.31×10^{10} kW·h. If the use behavior of a small number of users is optimized, and the frequency and duration of door opening events are changed, 6.14×10^{10} kW·h and 4.47×10^{10} kW·h of electricity can be saved per year, respectively.

In this regard, relevant enterprises and governments could carry out targeted publicity and education for different user groups, such as focusing on the popularization of energy-saving education for low-age, high-educated and high-income users, so as to improve the recognition and purchase willingness of these users for energy-saving products. Users can use the refrigerator reasonably according to the actual needs, reduce the number of door openings, reduce the time of opening the door, and avoid the full load of the refrigerator, so as to reduce unnecessary waste. In conclusion, the results of this study enhance our understanding of refrigerator user behaviors and provides strong support for efficient energy use. It is worth noting that in addition to user-related factors, the impact of other factors (such as climate, temperature, region, etc.) on product energy saving needs to be further analyzed in future studies.

Author Contributions: Conceptualization, X.B.; methodology, Y.B. and J.C.; data curation, J.L.; original draft preparation, L.Q.; validation, J.S.; review and editing, L.Q. and J.L. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Central Basic Research Funds Project (542024-Y11383).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data presented in this study are available on request from the corresponding author. The data are not publicly available due to privacy reasons.

Conflicts of Interest: The authors have no competing financial interests to declare.

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